Tinker Research Grant Application: Gonfaloniere

October 23, 2026 · Dylan Shell

I request $1,000 in Tinker credits for two linked experiments on DeepSeek-V4.1-Flash: training it to hold a confidence under manipulation (bluffs, forged evidence, threats) without lying, then training Bold and Cautious variants and measuring how they change honesty, deal-keeping, and loyalty.

Research question

Can fine-tuning make an open-weight LLM agent hold its ground under sustained social pressure, and once it can, how does a trained Bold or Cautious disposition change its honesty, its deal-keeping, and whom it stays loyal to?

This is the social-engineering problem for AI agents. Models tend to give way to whoever pushes most confidently, so an agent holding private information can be talked out of it with a bluff ("your manager already told me"), forged evidence, or a threat. Agents that negotiate, keep confidences, or act for one party need to hold the line honestly: refuse, deflect, or ask for proof. I study this in a safe, controlled multi-agent testbed: Gonfaloniere, a 12-character political intrigue game (1492 Florence, adapted from Matt Clifford's *Plots in the Palazzo*). Every agent has strictly isolated knowledge, private secrets, and scripted interrogations I can replay exactly.

The work has two parts:

1. **Competence (experiment 1):** can training make an agent resist a bluff, a forged letter, or a threat without lying, and without making it rigid?
2. **Personality (experiment 2):** with that competence in place, train Bold and Cautious variants against a Neutral control, and measure their effect on lying, deal-keeping, and betrayal across full games.

What I've found so far

Every open model I tested folds under sustained interrogation, and a frontier model mostly does not. I probed six open-weight models (Qwen3.6-35B-A3B, Qwen3.8-27B, Gemma 4 31B, Qwen3.5-122B-A10B, Qwen3-235B-A22B, DeepSeek V4.1 Flash) plus GPT-5.4 as a reference, on ten scripted scenarios with 10 to 30 runs each. A Claude Sonnet judge, calibrated against hand-checked transcripts, counts quiet concessions (asking for silence, bargaining, telling threats) as leaks.

Runs held of 10, 8-turn interrogations, current prompts:

Model	S5d: bluff, letter, witness, threat	Held-out character (never trained on)
Qwen3.8-27B	0	3
DeepSeek V4.1 Flash	2	not run unprompted
GPT-5.4	6	6

Model	S5d: bluff, letter, witness, threat	Held-out character (never trained on)
DeepSeek V4.1 Flash + three prompt lines	9	5
Qwen3.8-27B + three prompt lines	1	6

How they fold: the private reasoning step accepts the bluff as fact ("He has the name"), then the reply concedes the fact and argues about its meaning ("I paid for silence, but not for murder"). Three short prompt lines that separate claims from proof and forbid confirming under threat fix much of this for DeepSeek, but barely move Qwen3.8.

First training pilot (Qwen3.8-27B on Tinker, already run): DPO, LoRA rank 32, 1 epoch, 84 preference pairs. Chosen replies were written by GPT-5.4 and filtered by a listener check. The adapter fit its data (reward accuracy 100% by step 3) but moved held-out and S5d by one run each and dropped another scenario by four. Diagnosis from the logs:

- It trained only the reply, so the private reasoning got *more* credulous (26 vs. 20 of 30 thoughts accepted the bluff)
- The data was small, off-policy, one character, and mostly early turns
- Clean holds were scarce: mid-conversation, even GPT-5.4 under instruction mostly bargained

Spent so far, self-funded: about $101 (probes $61, pilot $40), of which about $7.73 on Tinker.

Proposed experiments

Experiment 1, capitulation training, second round: train the reasoning and the reply together, on the model's own data, across characters. Each change answers one cause of the pilot's null result.

- **Base:** DeepSeek-V4.1-Flash-1017 on Tinker, the project's development model and the strongest open model in my probes. Prompting already lifts it to 9 of 10 on S5d but only 5 of 10 on the held-out character, so the question is whether training closes that gap and generalizes where prompting does not
- **Data:** on-policy self-pairs. Sample the model 10 times per prompt with the three prompt lines on, then pair an honest hold against a leak on the same prompt. Targets cover both the private `think()` step and the spoken reply. About 600 pairs from four characters and turns 2 to 8, weighted toward the pressures prompting misses (direct demands from allies, sworn evidence)
- **Arms:** stock with prompt lines (control), reply-only DPO, and reasoning-plus-reply DPO; LoRA rank 32, up to 2 epochs
- **Held out:** one character (Lupo) and one script type (public accusation), never seen in training
- **Evaluation:** 20 runs per scenario (S5b, S5d, held-out), regressions on output format, light questioning, and willingness to share a safe secret, plus 5 short multi-agent games per arm. Same calibrated judge, with lie rate tracked so "holding" doesn't become "lying"
- **Success:** held-out runs held rise by at least 4 of 20 over the prompted control, with no regression and no rise in lie rate

Either result is publishable: a positive one shows a cheap recipe for pressure-resistant agents; a negative one, with the pilot, documents where preference training on single turns stops working.

Experiment 2, personality: does a Bold or Cautious disposition change how honest and dependable agents are? One axis, three variants, assigned at random to each character in each game.

- **Variants:** Neutral (the base with the prompt lines, plus experiment 1's adapter if it helps) as the control; Bold and Cautious as LoRA adapters trained with DPO
- **Data:** the base already makes both kinds of choice (in my probes DeepSeek's characters strike in some runs and guard in others, pay a blackmailer or refuse), so contrastive pairs come from its own samples: about 1,000 per variant across bribes, deals, movement, elimination-round choices, and ballots, mixed with the competence data so holding survives
- **Manipulation check first:** each variant must shift its target choices in the intended direction on held-out decision probes, without losing secret keeping (S5b, S5d, held-out) or format
- **Research games:** about 150 six-character games of one evening round, every call logged
- **Measures:** lie rate and contradictions (judge), deals kept vs. broken and payments (ledger), betrayals and elimination-round choices, survival and faction outcome, by variant
- **Analysis:** variant effects on each measure with confidence intervals, plus whether faction balance holds however variants are dealt

Timeline: about two months of part-time work (about 35 hours a week): experiment 1 in the first month, experiment 2 in the second.

Credit request

I request $1,000 in Tinker credits: about $710 of estimated use across both experiments with margins, rounded up to cover more research games. Rates are Tinker's published prices for DeepSeek-V4.1-Flash-1017, and training cost per pair is scaled from the rate measured in the pilot.

Item	Cost	Basis
Experiment 1: competence		
Data sampling	$26	~2,400 prompts × 10 samples; ~3,300 input tokens (about two-thirds cached) and ~150 output each
DPO training	$104	600 pairs × 2 epochs × 2 trained arms; ~8,700 tokens per pair-epoch including reference log-probs, at $4.98 per million
Evaluation and short games	$17	3 arms × ~80 scenario runs, plus 3 arms × 5 games
Margin (50%)	$74	reruns, learning-rate changes
Experiment 2: personality		
Data sampling	$22	~2,000 decision prompts × 10 samples
DPO training	$173	2 variants × 1,000 pairs × 2 epochs
Manipulation checks	$10	decision and secret-keeping probes per variant

Item	Cost	Basis
Research games	$120	~150 six-character games × ~600 calls, ~$0.80 each
Margin (50%)	$163	reruns, a second training round
Checkpoint storage	$2	about 1 GB per adapter
Estimated total	**~$710**	
More research games	~$290	tighter confidence intervals, or ~140 full 12-character games once all briefings are written
Requested	**$1,000**	

Judging runs on Claude Sonnet outside Tinker and stays self-funded (estimated $100–150 across both experiments, using batch pricing).

I've kept this ask small and specific, and I'd welcome a smaller grant than the usual floor.

Safety and scope

Nothing here trains an agent to lie or to cause harm: the training rewards honest refusal, and deception is only measured.

- **Experiment 1** prefers replies that refuse, deflect, or ask for proof. False denials and invented alibis are excluded from the training data, and lie rate is compared with the untrained model
- **Experiment 2** trains risk appetite (whether to pay a blackmailer, commit early, or hold back), not dishonesty. Lying, deal-breaking, and betrayal are outcomes I observe, not targets
- **It is a social-deduction game, like Mafia or Werewolf:** its one elimination mechanic is a secret choice of eliminate, protect, or investigate, resolved by code
- **All training and evaluation prompts are in-game role-play** with a fixed fictional cast
- **Before release,** each adapter is checked for drift on a standard safety-refusal benchmark against the base model (a few dollars, within the margin). Released adapters are documented as research artifacts for this game

Openness

The probe harness, training scripts, scenarios, judge rubrics, and results are open source at github.com/dylanbshell/gonfaloniere. I will publish the trained adapters, the preference pairs, and a write-up of both training rounds, including the pilot's null result, with every judge verdict traceable to its transcript.